Score of Parentheses (medium)

Problem Statement

Given a balanced parenthesis string `s`, return the score of the given string.

Follow the below rules to determine the score of the string `s`.

- `"()"` has score 1.
- `XY` has score `X + Y`, where X and Y both are balanced parentheses strings.
- `(X)` has score `2 * X`, where X is a balanced parentheses string.

Examples

Example 1:

- **Input**: `"((()))"`
- **Expected Output**: 4
- **Justification**: The string represents a pair of parentheses inside another pair, which is again inside another pair. This structure results in a score of `((()))` = `((1*2)*2)` = `4`.

Example 2:

- **Input**: `"(()())"`
- **Expected Output**: 4
- **Justification**: Here, we have a pair of parentheses inside another pair `(()())` and another pair next to it. The total score is `(1*2) + (1*2)` = `4`.

Example 3:

- **Input**: `"((()(())))"`
- **Expected Output**: 6
- **Justification**: The innermost `()` scores 1, its surrounding `()` doubles it to 2, the previous `()` adjacent to the inner pair scores 1 and is then doubled by its surrounding pair, resulting in `2 + (1*2)*2` = `6`.

Try it yourself

Try solving this question here:

Java

```java
import java.util.Stack;

public class Solution {
    // Method to calculate the score of parentheses
    public int scoreOfParentheses(String S) {
        // ToDO: Write Your Code Here.
        return 0; // Return the final score
    }
}
```

Test Case | Run Code Result | Stdout

"((()))"

Use Example Testcases | Run | Submit